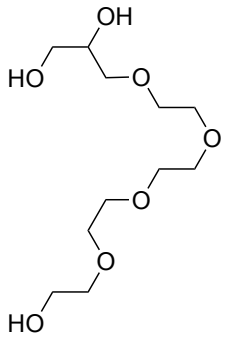
3.2.S.1.2 Structure [2439-exNA, Hongene]

PrP-lowering divalent siRNA 2439-exNA (2439-exNA, in abbreviate in the rest of this document) comprises the oligonucleotide duplex structure formed by the association of 1 equivalent of 2439-exNAS (sense strand) and 2 equivalents of 2439-exNAA (antisense strand) through specific Watson-Crick-Franklin base pairing. The sequence identity of 2439-exNAS and 2439-exNAA are shown in Table 1.

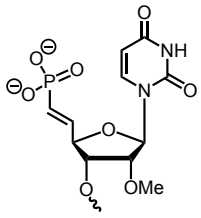
Table 1: Sequences of component strands of 2439-exNA drug substance

Strand Name	Strand Sequence
Antisense (Hongene code; OS121)	5'-V(mU)#(fG)#(mA)(fA)(fU)(fA)(mC)(fU)(mC)(fA)(mC)(fA)(mA)(fA)(mG)(fU)(mG)(mC)(mA)#(mxU)#(fxU)-3'
Sense (Hongene code; OS122)	5'-(mA)#(mC)#(mU)(fU)(mU)(fG)(mU)(fG)(mA)(fG)(mU)(mA)(mU)(fU)#(mC)#(mA)-3'-DIO-3'-(mA)#(mC)#(fU)(mU)(mA)(mU)(fG)(mA)(fG)(mU)(fG)(mU)(fU)(mU)#(mC)#(mA)-5'

Notes:
mA, mC, mG, mU – 2′-*O*-methyl nucleotides
fA, fC, fG, fU – 2′-fluoro-2′-deoxy nucleotides
– Phosphorothioate linkages

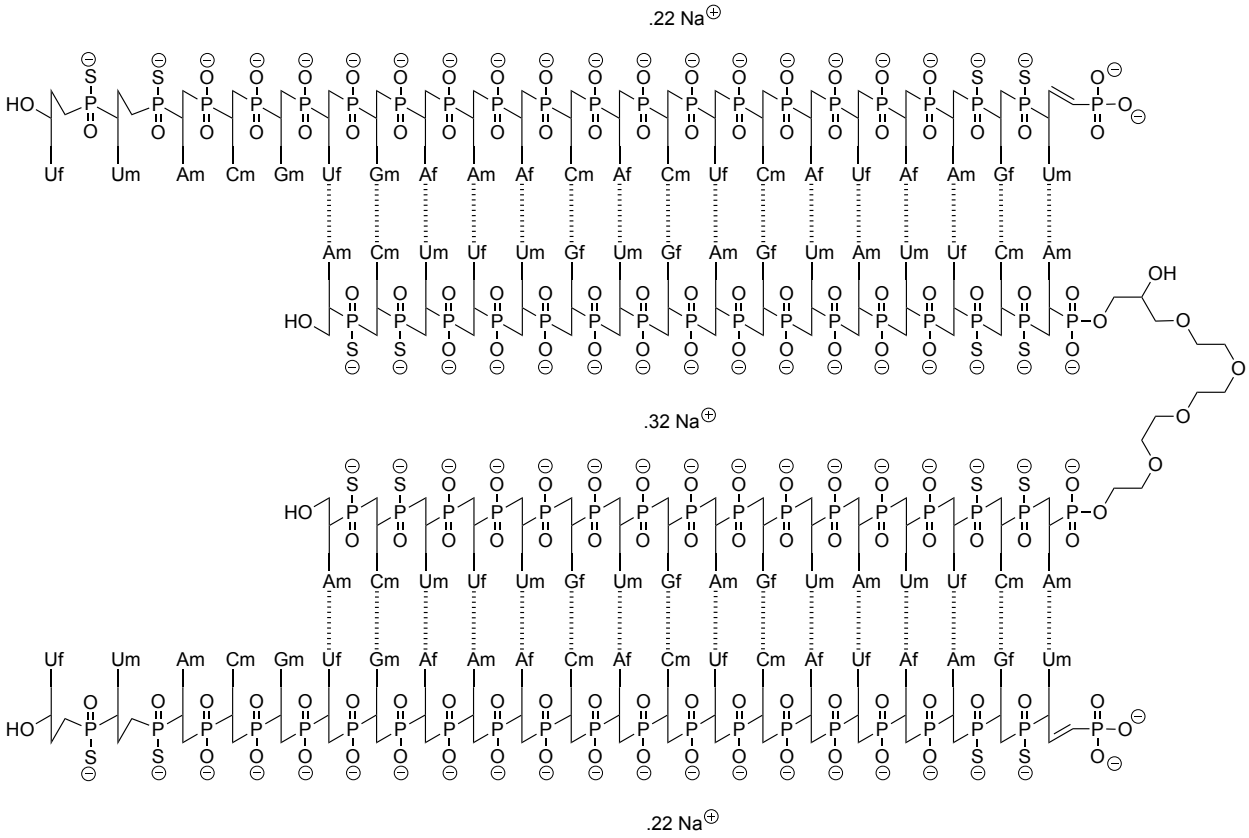
DIO - 

tetraethylene glycol glycerol linker

V(mU) - 

2439-exNA triplex drug substance contains 2 moles of CP-OS121 and 1 mole of CP-OS122. Alternatively, the structure of 2439-exNA, showing how the 2 antisense strands (OS121) and the 1 sense strand (OS-122) can also be represented after hybridization by the graphic depicted in Figure 1.

Figure 1: 2439-exNA Sequence



The 2439-exNA drug substance is a hybridized oligonucleotide containing 2 equivalents of antisense strand and 1 equivalent of sense strand (see figure 1 above).

The chemical structures of 2439-exNAS and 2439-exNAA are shown in Figure 2 and Figure 3, respectively.

Figure 2: Structure of sense strand

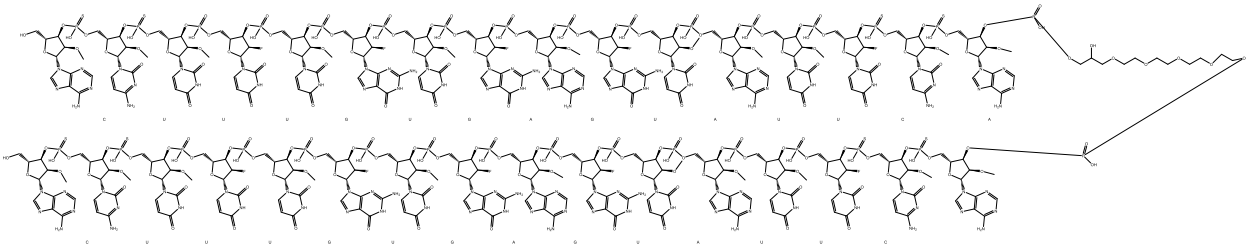
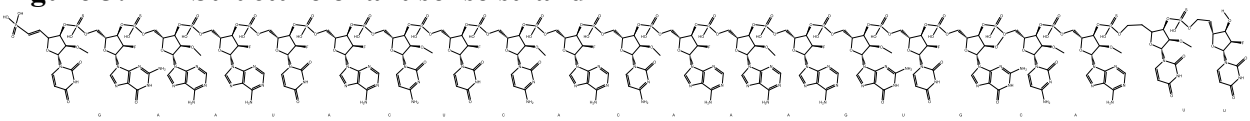


Figure 3: Structure of antisense strand



3.2.S.1.2.1 Molecular Formula and Average Molecular Mass

The molecular formula and relative molecular mass of 2439-exNA drug substance, the sense and antisense strands are provided in Table 2.

Table 2: Molecular Formula and Average Molecular Mass of 2439-exNA

	2439-exNA (duplex) **2 equivalents antisense + 1 equivalent sense**	**antisense**	**sense**
Molecular formula (sodium salt)	$C_{763}H_{880}F_{30}N_{268}O_{481}P_{74}S_{16}Na_{76}$	$C_{214}H_{242}F_{10}N_{79}O_{131}P_{21}S_4Na_{22}$	$C_{335}H_{396}F_{10}N_{110}O_{219}P_{32}S_8Na_{32}$
Molecular formula (free acid)	$C_{763}H_{956}F_{30}N_{268}O_{481}P_{74}S_{16}Na_{76}$	$C_{214}H_{264}F_{10}N_{79}O_{131}P_{21}S_4$	$C_{335}H_{428}F_{10}N_{110}O_{219}P_{32}S_8$
Molecular weight (sodium salt)	26531.16 Da	7445.21Da	11640.74 Da
Molecular weight (free acid)	24952.48 Da	7007.59 Da	10937.31 Da

3.2.S.1.2.2 Stereochemistry

It is known that the ratio between Rp and Sp stereoisomers in each PS linkage depends on the activator used. The experiments described in the paper by Ravikumar & Cole (2002) show how the Rp/Sp ratio of the PS linkage is reproducible, within limits of experimental error, between syntheses and is independent of diastereomeric composition of phosphoramidites, synthesis scale, solid support type, reactor, concentration during coupling or synthesis conditions.

The sense strand contains 8 phosphorothioate (PS) linkages and the antisense strand contains 4 phosphorothioate (PS) linkages. Due to the process used for the coupling (chain elongation) step during oligonucleotide synthesis each PS can exist in the Rp or Sp form. The antisense strand contains 2^4 = (16) stereoisomers, the sense strand contains 2^8 = (256) stereoisomers and 2439-exNA duplex drug substance 2^{12} = 4096 stereoisomers.

All ribose sugars that are constituent in the nucleotides are the naturally occurring D-ribose form and are stereopure. No chemical reactions in the manufacture of 2439-exNA occur at these stereocenters and thus they are unchanged.

At the terminus of the antisense strand is a nucleoside phosphonate isostere. This nucleoside phosphonate contains an alkene functionality in the “E” configuration.